HYPHAE
COMPOSITES

Buy less new resin.
Transform your scrap.

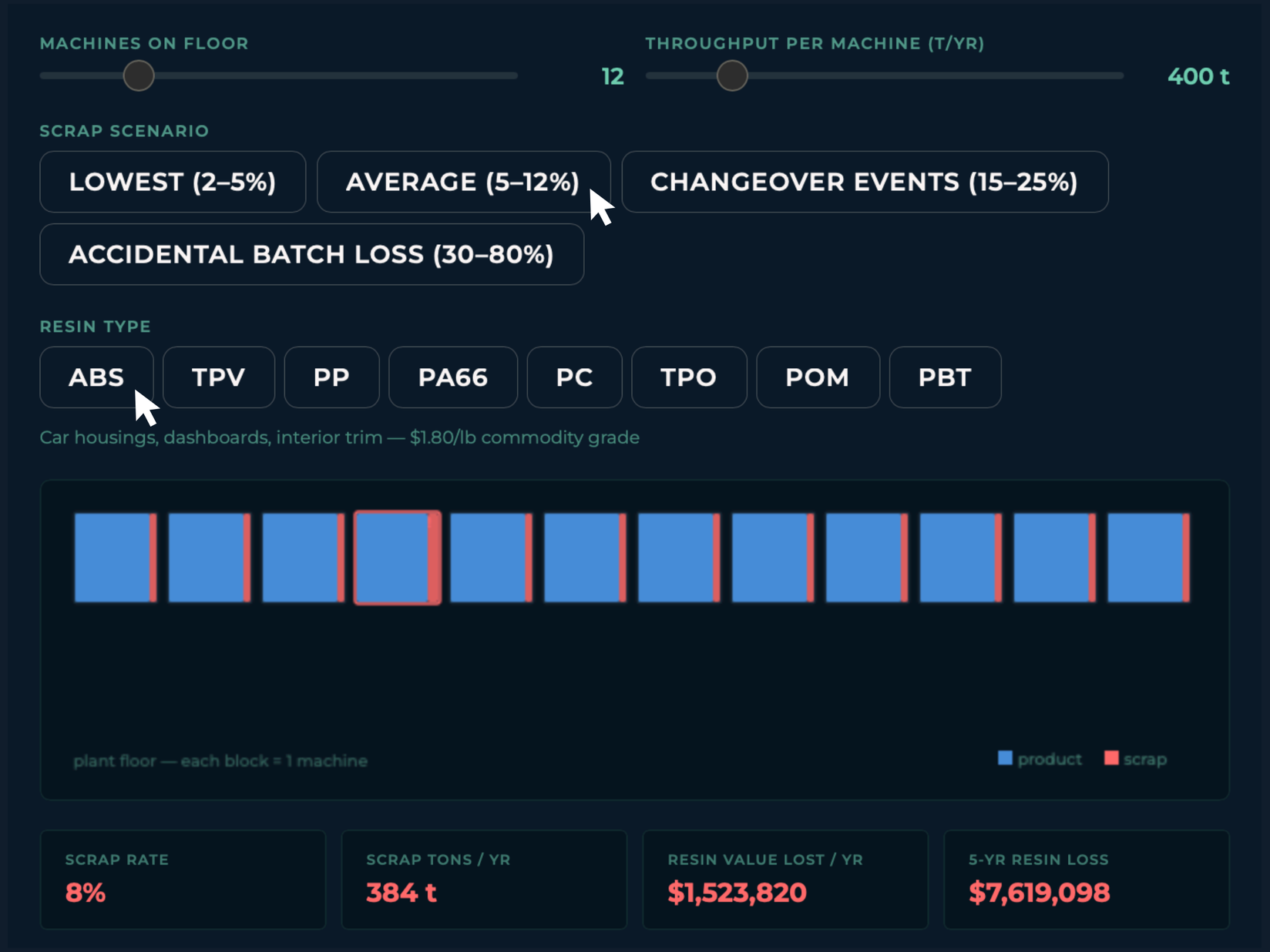
HYPHAEC.TECH

What is inside this deck?

1. **The Problem: A Simulator Sample of the Value Lost As Scrap In a Plant**
2. **How Do Our Customers Handle Scrap Today?**
3. **What is Hyphae? & How do we solve the problem**
4. **Our Business Model: Financial + Environmental Value**
5. **Get to know The Team**
6. **Our Moat and Who Else is Playing?**
7. **Next Steps**
8. **Contact Us**

The Value Within *An Extrusion Plant Simulator*

**Can be applicable to injection molding and industrial 3D printing farms.*



This is a simulator of a Tier 2 supplier facility that makes car housings from ABS (plastic). We selected 12 machines for this medium-sized plant.

Extruders are shown as blue rectangles and in red is the "proportion" of scrap produced for better visualization.

If each machine makes the average percentage of scrap (wide range), in this case 8%. The facility looses over $1.5M/year only counting material value.

Current Options

To deal with plastic scrap and production failures do not often bring value back to the manufacturers.

OPTION 01
Landfill

- Material value permanently destroyed
- Tipping fee $78/ton — you pay to lose it
- Logistics and waste handling on top
- ESG liability growing every year

OPTION 02
Sell to recyclers

- Cannot handle advanced blends — TPVs, coextrusions
- Rejected or downcycled — no automotive grade output
- You lose the material and receive pennies
- Supply chain loop never closes

OPTION 03
Regrind

- Heat and shear already broke the molecular chains
- Regrinding breaks them further
- Fails automotive spec — every time
- Downcycled at best — low value applications

PRODUCT

A physical AI unit that repairs plastics at the molecular level

Something that has never been proven at the commercial level.

PREVIOUS EXPERIMENTS AT THE BENCH SCALE:



Average of 94% score recovery of physical properties among different plastic types such as TPV, Fiberglass, PETg, ABS.
Technique used: reactive extrusion

For co-extruded or mixed compounds our recovery score is of 92% for Santoprene (TPV).
Technique used: compatibilization.

BETA VERSION INTEGRATED ON OCT 2026



#64/082,024
PROVISIONAL PATENT
hardware unit

BETA version top features:

- *Stable 2mm pellets as output*
- *Automated*
- *Can perform plastic recovery between 94-97%*
- *Operator assisted*
- *Wi-fi connected*
- *1m x 1m floor footprint*

ALPHA version (To be unlocked after first pilot results and pre-seed round Q3 2027)
ttop features:

- *Can switch output between pellets and filament*
- *Data gathering*
- *Masterbatch real-time dosage adjustment (ML)*
- *Stable recovery of 99%*
- *Standalone*

Our *Hardware-as-a-service*

WITH HYPHAE
closed loop on your floor



↺ closed loop

SCRAP PROCESSED / MO
32 t

RESIN VALUE RECOVERED / MO
$126,985

HYPHAE MONTHLY COST
$19,180

CO_2 AVOIDED / MO
67 t avoided

LEASE $11,500 + MASTERBATCH $240/TON
$107,805 / mo
net recovery after all Hyphae costs

**Same simulator data*



Given the manufacturing plant scenario of Page #3

- *Hyphae can save our customers over $1M per year.*
- *Hyphae can earn an average of $228k per year from each on-site unit.*
- *Avoiding approximately 67 CO2 tons per month related to the making of new plastic resin and diverting the equivalent of tons reprocessed from the landfill.*

$0 Capex
HARDWARE LEASED
no facility investment to start

$107,805	$1,293,660	6.6x	Day 5
NET GAIN / MONTH	NET GAIN / YEAR	ROI ON HYPHAE COST	PAYBACK PERIOD

WHO IS BUILDING? Here is a little bit about our team



David Esparza

CTO & CO-FOUNDER

Managed a small injection molding facility and took it from 0 to thousands in revenue per month.
He leads technical innovation, team coordination, and AI integration at Hyphae.

- He believes in revolutionizing American manufacturing through affordable, self controlling systems that minimize waste, maximize efficiency, and make advanced industrial technology accessible to businesses of all sizes.



Adriana Esparza

CEO & CO-FOUNDER

Worked in biotechnology and entrepreneurship is her way to live.
She pitches, brings the team together, and leads vision at Hyphae.

- She believes in the power of innovation as our best tool to enhance our industries and with it, our lives.



David E.Gonzalez

DIRECTOR OF ENGINEERING

Led large teams across North America for over 25+ years in the automotive industry.
He's been recognized for boosting innovation, productivity, and complex problem solving.

He leads automation, integration, and development at Hyphae



Michael Gonzalez

CMO

Has led marketing campaigns among big brands such as Wings Army and the Dallas Fort Worth Airport.

He leads the marketing strategy and branding at Hyphae.



Francisco García

ENGINEERING MANAGER

Has performed over 5 thousand testing equipment calibrations, material and equipment testings.

He runs our production-line for Hyphae's recurring customers.



Juan Torres

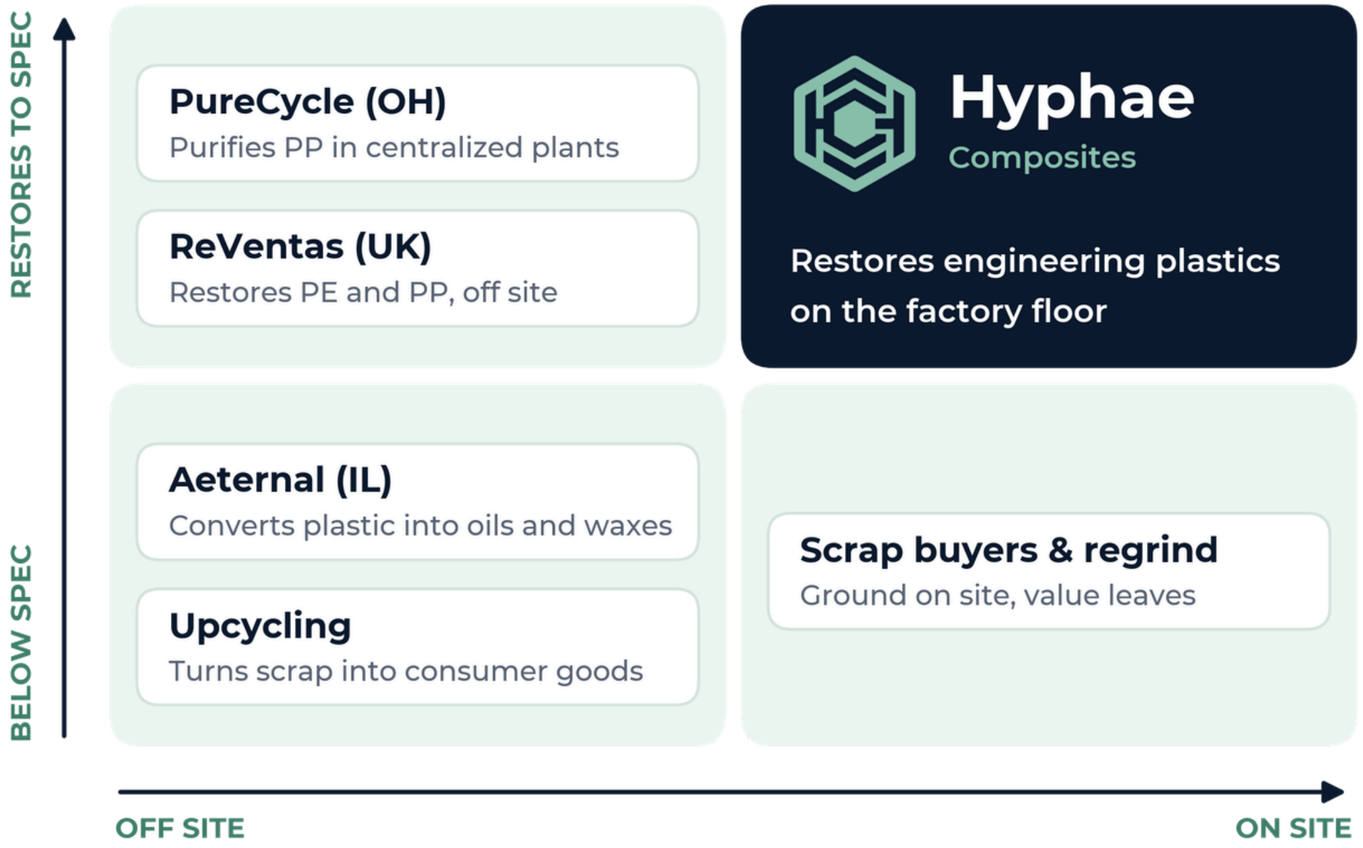
JUNIOR ENGINEER

Our most junior. Learns and delivers quickly while finishing his Eng. degree.

He has programmed more than 5 Hyphae unit features and builds electronic components at Hyphae.

COMPETITION

We all serve this mission.



Our Moat

[We know our plastics and our customers]

- **Trade-secret chemistry.** Hard to copy today and adaptive to every run. *Solves for plastic properties.
- **Data that compounds.** Every unit learns from every batch it restores. The more plants we're in, the smarter every unit gets. *Solves for plastic variability and complex mixes.
- **Built into the plant.** Once our unit runs on their floor, it's part of their production-line, proving recurring value. *Solves for operational technology (OT) and industry 4.0

ALSO IN THE SPACE

HOPPER by plus10 prevents scrap but doesn't recover it. BASF IrgaCycle sells additives, not a system.

Thinking of hosting a pilot?

See how much value we can bring by using our simulator:

https://hyphaecomposites.github.io/plant-simulator/

Thinking of participating in our pre-seed round?

Join our investors on Q3 2027

HYPHAE
COMPOSITES

E-mail Us At:

connect@hyphaec.tech

Or Visit Us At:

https://hyphaec.tech